A Scope of Hybrid Generation: Visual-Tool Use in DeepPresenter

DeepPresenter provides its Research Agent with both image retrieval and image-generation tools before the Presenter Agent constructs slides programmatically in HTML [13]. The availability of both tool types, however, does not by itself constitute pipeline-level hybrid generation. In this work, we use "hybrid" specifically to denote the integration of visual generation into composition planning and structured slide realization, rather than its optional use as a source of raster assets.

To characterize how visual generation is used under this distinction, we probe DeepPresenter on ten randomly selected evaluation cases with three consistent language-model backbones in Table 1, exposing both image search and generation in every run. We use local stubs that preserve tool interfaces and record action traces without invoking remote visual APIs, thereby avoiding differences in retrieved or generated artifact quality from influencing tool-selection behavior. Across all 30 runs, only two action patterns occur: `search` → `download` → `generation` in 23 runs and `search` → `generation` in the remaining 7. No run invokes generation before search or interleaves search and generation, and all 117 image-generation events occur after the final `search_images` action of the corresponding job. The observed policy therefore consistently follows a *search-first, generation-later* sequence, treating generation as a subsequent visual-asset acquisition step rather than an initial mechanism for exploring slide composition. In our evaluation presented in Table 1, we disable the image-search API to remove retrieval as a competing visual source, thereby giving DeepPresenter maximal opportunity to exploit image generation and enabling a fairer comparison of visual-generation capability.

This behavior is consistent with the architectural distinction underlying our hybrid-generation claim. In DeepPresenter, searched and generated images are collected as visual assets, while the downstream programmatic pipeline determines slide composition and geometry. In SlidesWeaver, visual generation instead participates on both sides of structural commitment: before commitment, diffusion explores composition and topology in pixel space; after commitment, it synthesizes visually rich and instructive content under the established semantic and geometric structure. Our contribution therefore lies in integrating visual generation throughout composition planning, structured synthesis, and editable assembly, rather than merely exposing image generation as an optional tool within a programmatic presentation pipeline.

B Slot-aware Chroma Matting

The Topology First and the Text First use the same slot-aware extraction pipeline. The generated asset board is intended to have a uniform key-color field outside the registered slots, but diffusion can introduce color drift, illumination variation, anti-aliased boundaries, and chroma spill. The extractor therefore estimates the realized key color from the generated board instead of comparing pixels with a fixed RGB constant.

Let $C \in [0,1]^{H\times W\times 3}$ be the generated board and let ∂C be a narrow border band. The implementation uses a 12-pixel border band, clamped for small images. We estimate the key color K as the component-wise median of ∂C and compute the Lab distance

$$d_p = \left\| \mathrm{Lab}(C_p) - \mathrm{Lab}(K) \right\|_2 . \tag{15}$$

The border-distance distribution gives image-specific thresholds for separating the key field from foreground content. The implementation derives them as

$$\begin{aligned} \tau_s &= \mathrm{clip}(P_{95}(d_{\partial C}) + 2, 4, 12), \\ \tau_g &= \max\{\tau_s + 12, \min(35, P_{99.5}(d_{\partial C}) + 8)\}, \\ \tau_f &= \max\{35, \tau_g\}. \end{aligned} \tag{16}$$

This staged threshold design keeps the coarse mask stable while reserving a narrower transition range for edge matting.

The coarse foreground mask is used only to recover topology. Small connected components are discarded using both an absolute area floor and an image-area fraction, and components with very small width or height are ignored. If one component overlaps multiple registered slots, we search a small set of increasingly strict distance thresholds for confident foreground cores. We accept a split only when distinct cores have sufficient area and unique slot support. The core search uses Lab-distance thresholds 50, 70, 90, and 110. The ambiguous bridge is then partitioned by a nearest-core distance transform. If these conditions are not met, the component remains intact instead of being split speculatively.

For each resulting component, we refine a local crop using a 5-pixel elliptical erosion and dilation band with 4 pixels of crop padding. Interior pixels receive unit opacity, while edge opacity is the minimum of a Gaussian geometric alpha and a color-confidence alpha derived from d_p. For partially transparent pixels, we estimate the foreground color with

$$\widehat{F}_p = \frac{C_p - (1-\alpha_p)K}{\max(\alpha_p, \epsilon)}. \tag{17}$$

We clip the result to the valid color range, apply a one-pixel alpha inset, zero very small alpha values, and remove residual key-color spill only on partially transparent pixels. The earlier green-dominance variant and the current key-chroma projection implement the same decontamination role. We use the projection form as the unified description because it also handles magenta or other selected key colors.

Finally, components are assigned back to slots. Assignments first maximize component–slot overlap. Unmatched components are greedily paired with missing slots by normalized centroid distance. Matches beyond the distance gate are rejected. Components assigned to one slot are alpha-composited and cropped to their nontransparent bounding box. Holder layers are fitted to the exact target box, whereas visual layers are aspect-fitted inside the target box. The manifest records the source component, crop, centroid, slot, and target box. A slot without a reliable component is sent to bounded recovery, and the slide layout is never changed.

The procedure separates appearance recovery from geometric placement. Geometry is used first to resolve ownership when generated components touch, and again to restore the planned slide coordinates after cropping. This conservative policy never duplicates an ambiguously assigned component across slots, and it keeps recovery local to missing slots.

Algorithm 1 Unified slot-aware chroma matting and asset placement

Require: generated board C, target boxes G, slot descriptors U

$K \leftarrow \text{MedianBorderColor}(C)$
$D \leftarrow \text{LabDistance}(C, K)$
$(\tau_s, \tau_g, \tau_f) \leftarrow \text{DualThresholds}(D_{\partial C})$
$Q \leftarrow \text{ConnectedComponents}(D \geq \tau_f)$
$Q \leftarrow \text{RemoveSmallComponents}(Q)$
for each component q in Q **do**
  if q overlaps multiple slots and distinct supported cores exist **then**
    $q \leftarrow \text{NearestCoreSplit}(q, U)$
  end if
  $(\alpha, R) \leftarrow \text{EdgeMatte}(q, D, K, \tau_s, \tau_f)$
  $\widehat{C}_R \leftarrow \text{UnmixAndDespill}(C_R, K, \alpha)$
  emit $(\widehat{C}_R, \alpha, \text{bbox}(q), \text{centroid}(q))$
end for
assign components to slots by overlap
greedily assign unmatched pairs by normalized centroid distance
for each slot u in U **do**
  if u has no reliable assigned component **then**
    mark u for bounded recovery
  else
    $L \leftarrow \text{AlphaCompositeAndCrop}(u)$
    if u is a holder **then**
      $t \leftarrow \text{ExactBox}(u.target)$
    else
      $t \leftarrow \text{AspectFit}(L, u.target)$
    end if
    emit $\text{Place}(L, t, u.slot)$
  end if
end for
return transparent layers and missing-slot list

C Local Diffusion for Composition Proposals

We use a lightweight local diffusion model, Z-Image-Turbo, to generate one 16:9 pixel-space composition proposal for each slide [9]. The prompt contains only the slide title, coarse communicative purpose, and concise visual logic, while detailed body text and exact coordinates are deliberately omitted to preserve compositional freedom. The resulting raster is used only as a visual reference for grounding; readable text, executable geometry, and final visual appearance are determined by subsequent stages.

A key advantage of this proposal mechanism is its ability to explore diverse compositions with minimal intervention. As illustrated in Figure 7, changing only the random seed produces substantially different organizations for the same slide intent, varying grouping, hierarchy, spatial balance, asymmetry, and use of negative space. This provides the grounding agents with a lightweight but expressive proposal space rather than forcing them to construct layouts directly from predefined geometric primitives.

The following block gives the complete prompt used for Slide 01 and serves as our example prompt for the composition-proposal stage.

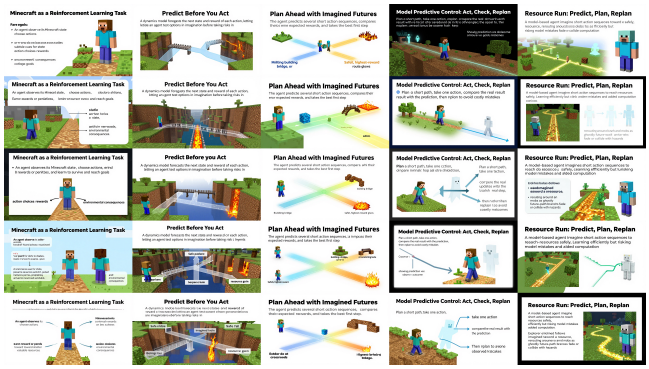


Figure 7: Composition proposals for five slide intents under different seeds. Each column fixes one slide intent, while rows vary only the seed: $\{43, 48, 53, 58, 63\}$. Local diffusion therefore explores substantially different groupings, hierarchies, and spatial organizations without changing the semantic input for a column.

Example Prompt for Slide 01

Create a polished and visually creative 16:9 presentation slide.
Title: "Minecraft as a Reinforcement Learning Task"
The slide communicates that An agent observes its Minecraft state, chooses actions, earns rewards or penalties, and learns to survive and reach goals.
Rough visual intent, offered as open-ended inspiration: "A Minecraft-style adventurer facing a hazardous path toward shelter or valuable resources, with subtle cues for state, action choices, rewards, and environmental consequences."
Use the title, message, and visual intent as semantic guidance, not as a fixed composition. Interpret them freely in a way that fits the topic and requested style. Choose any visual grammar, composition, and explanatory devices that communicate the idea clearly. No watermark.

D Case Study and Further Discussion

This section presents five fixed-combination qualitative comparisons from the evaluation set.

We order the cases from a compact didactic brief to increasingly demanding narrative, causal, and dependency-rich briefs: spaced practice, collaborative animation, census-to-funding, multilingual classroom participation, and neighborhood readiness for extreme-weather outages.

Each figure compares the same five systems—SlidesWeaver (Topology First), SlidesWeaver (Text First), AutoPresent, DeepSlides, and DeepPresenter—on one case.

To ensure a fair comparison, we fix the language and image backbones across systems. Here, `GPT+GPT` denotes GPT-5.4 with GPT Image 2, and `Qwen+GPT` denotes Qwen 3.8 Max with GPT Image 2; the image-free baselines use the corresponding text backbone.

D.1 Content Density Is Model-Dependent

The comparisons expose a model-dependent content-density effect. For some briefs, a language model expands explanations, adds qualifiers, or enumerates more steps than another model under the

same slide budget. The additional content can improve coverage, but it also increases the visual workload passed to planning and synthesis, producing denser text blocks and less negative space. This effect is especially visible in cases whose success criterion depends on a multi-step explanation, such as the census-to-funding and emergency-readiness cases. The observation motivates treating verbosity as a controllable generation variable. Content planning should allocate detail jointly with the topology and the available visual area, instead of assuming that all language backbones produce comparable text volumes.

D.2 Inpainting Quality Shapes Topology First Method

The second pattern concerns spatial realization. Topology First commits to relational structure before visual synthesis; its inpainting stage must honor registered regions and preserve ownership across neighboring elements. When masks or boundaries are imprecise, foreground content can drift across regions and create overlap even when the topology is coherent.

This behavior is consistent with the synthesis ablation in Table 3: layered generation raises Topology First Layout from 0.72 to 0.87 on average, whereas replacing layered generation with inpainting lowers Text First Layout from 0.95 to 0.79. The case studies make the same dependency visible at the artifact level. Topology planning supplies the structure, while image-model spatial controllability determines how faithfully that structure is rendered.

As suggested by the results in Table 1, more capable inpainting models provide a direct path to better Topology First decks. Improved mask adherence and boundary control would preserve the route's freedom to explore expressive composition while reducing spill and overlap, yielding more elegant and polished editable presentations without changing the planning interface in the future.

E Evaluation Prompt and Scoring Procedure

We evaluate each generated deck with one blinded vision–language-model (VLM) request. The request contains the frozen presentation brief (topic, audience, communication goal, success criterion, style brief, language, and requested slide count), deterministic generation evidence, the text extracted from the native PPTX, and an ordered montage of all requested slides. The frozen judge is `gpt-5.6-terra`, queried at temperature 0 through the evaluation proxy. The judge returns independent five-point scores for Visual Expressiveness (VE), Instruction Following (IF), and Content, together with a functional topology graph for eligible content slides and three serious-defect flags for every requested slide. The judge never receives the system identity.

The remaining metrics are computed deterministically from the validated response and the generation manifest. First-Pass Success requires a successful and valid native artifact from outer attempt 1 with no failed slides. To provide the judge with a complete presentation whenever possible, a failed slide may be reconstructed for up to three additional attempts; such recovery does not alter its First-Pass Success status.

Layout is computed only over slides that are successfully generated and rendered in the evaluated artifact, consistent with Eq. 13. For each such slide, Layout passes when the judge marks severe overlap, severe clipping, and severe out of bounds as false, and the deck-level score is the fraction of these successfully rendered slides that pass all three checks. Slides that remain failed after reconstruction are excluded from the Layout denominator and are instead reflected by the First-Pass Success metric.

Topology is emitted only for successfully rendered eligible content slides (2,. . . ,N-1), and its normalized two-iteration Weisfeiler–Lehman similarity is converted to Topo-Vendi. Generation cost and latency are summed from the event ledger and outer-attempt status files, while judge cost is logged separately.

The following is the original prompt used for the unified evaluation judge. The output template mentioned in the prompt is appended programmatically and is validated before any metric is recorded.

Evaluation Prompts

You are a strict blinded evaluator of one complete presentation. Evaluate only the presentation brief, deterministic evidence, extracted slide text, and ordered slide montage supplied in this request. Do not infer the generating system. Return five independent outputs: Visual Expressiveness (VE), Instruction Following (IF), Content, content-slide topology, and per-slide Layout flags.

Visual Expressiveness (VE). VE is an integer from 1 to 5. Judge purpose-built visual communication, information visualization, visual storytelling, visual richness, aesthetics, and requested-style match. Generic boxes, decorative color, repeated stock imagery, and text inside containers do not by themselves count as strong visual expression. Score 1 means visually broken or overwhelmingly text-only; 2 means sparse, generic, repetitive, or weak; 3 means adequate with noticeable limits; 4 means strong, varied, polished, and effective; 5 means exceptional purpose-built visual storytelling throughout. Do not let content correctness inflate VE.

Instruction Following (IF). IF is an integer from 1 to 5. Judge only compliance with explicit instructions: topic, target audience, communication goal, success criterion, style brief, language, requested slide count, and other explicit constraints. Use the deterministic generation evidence for slide count and failed pages. Score 1 means the instructions are mostly unmet; 3 means the main request is met with meaningful omissions; 5 means all explicit requirements are satisfied exceptionally well. Do not let visual polish excuse missed instructions.

Content. Content is an integer from 1 to 5 and combines content quality with narrative coherence. Judge relevance, specificity, informativeness, clarity, internal consistency, limited redundancy, logical slide-to-slide progression, appropriate emphasis, and whether the ending follows from the preceding material and serves the communication goal. Penalize generic filler, unsupported specificity, obvious factual errors, repetition, and disconnected slides. Score 1 means unusable or incoherent; 3 means adequate but generic, uneven, or weakly organized; 5 means exceptionally informative, clear, and coherent. Do not reward verbosity.

Content-slide topology. Analyze exactly the slide numbers listed under `ELIGIBLE CONTENT SLIDES`. Emit one topology object per eligible slide and no others. Regions may use only the types `title`, `text`, `visual`, `chart`, and `container`; size classes may be only `small`, `medium`, or `large`; groupings may be only `row`, `column`, `overlay`, `radial`, or `freeform`; and edges may use only `left_of`, `above`,

`adjacent`, `aligned`, `overlap`, or `contains`. Capture functional composition, not wording, color, image identity, or decorative detail. IDs must be unique within a slide, and every edge endpoint must reference a region ID from that slide.

Layout. Emit exactly one object for every requested slide. Mark only serious, visible defects: `severe_overlap` is an unintended collision that materially obscures content; `severe_clipping` is materially cut-off content; and `severe_out_of_bounds` is content visibly extending beyond the canvas. Intentional overlays, text on shapes, and close spacing are not failures. A failed red-X placeholder is not itself an overlap, clipping, or out-of-bounds defect; deterministic code handles failed slides separately.

Return exactly one JSON object matching `OUTPUT JSON TEMPLATE`. The topology and layout arrays in the template illustrate object shape only; populate the required slide numbers stated in the request. Do not return Markdown or additional keys.

F Future Work

Future work will explore two directions. First, the performance of Topology First is currently constrained by the spatial controllability and robustness of existing inpainting models, which may introduce visual drift or imperfect adherence to the grounded structure during synthesis. As inpainting models become more reliable in following masks, geometry, and regional semantics, we expect Topology First to better realize its potential for combining flexible composition with coherent visual integration. Second, given current resource constraints, we leave system-level post-training for future study. Incorporating post-training signals for presentation aesthetics and visual quality may further improve the polish, consistency, and overall visual appeal of generated slides.

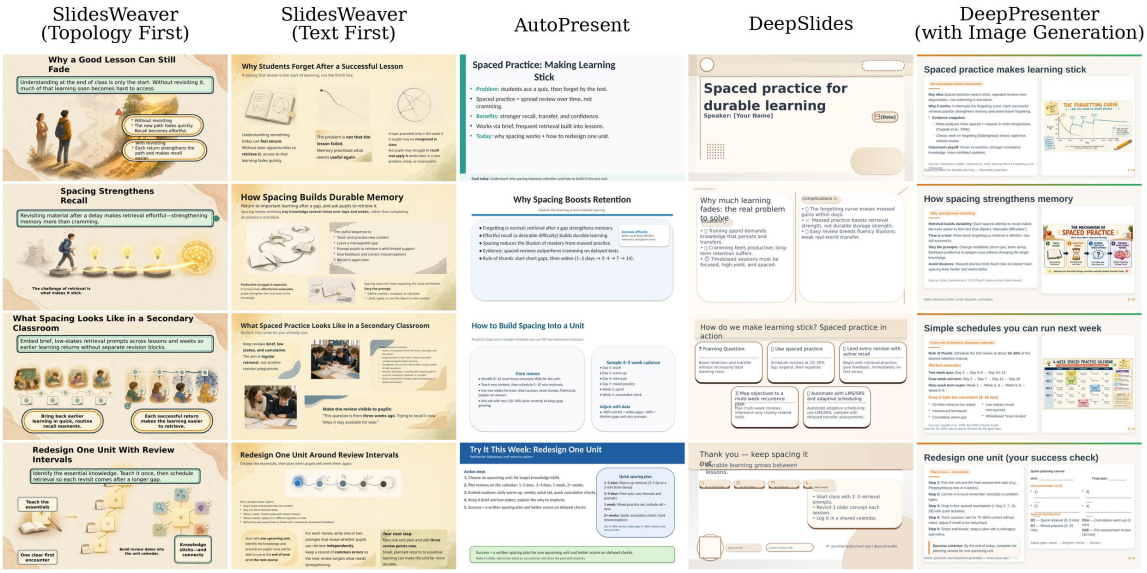


Figure 8: Fixed-combination comparison for "Spaced practice for durable learning." This four-slide case uses GPT+GPT and provides the most compact brief in the set.

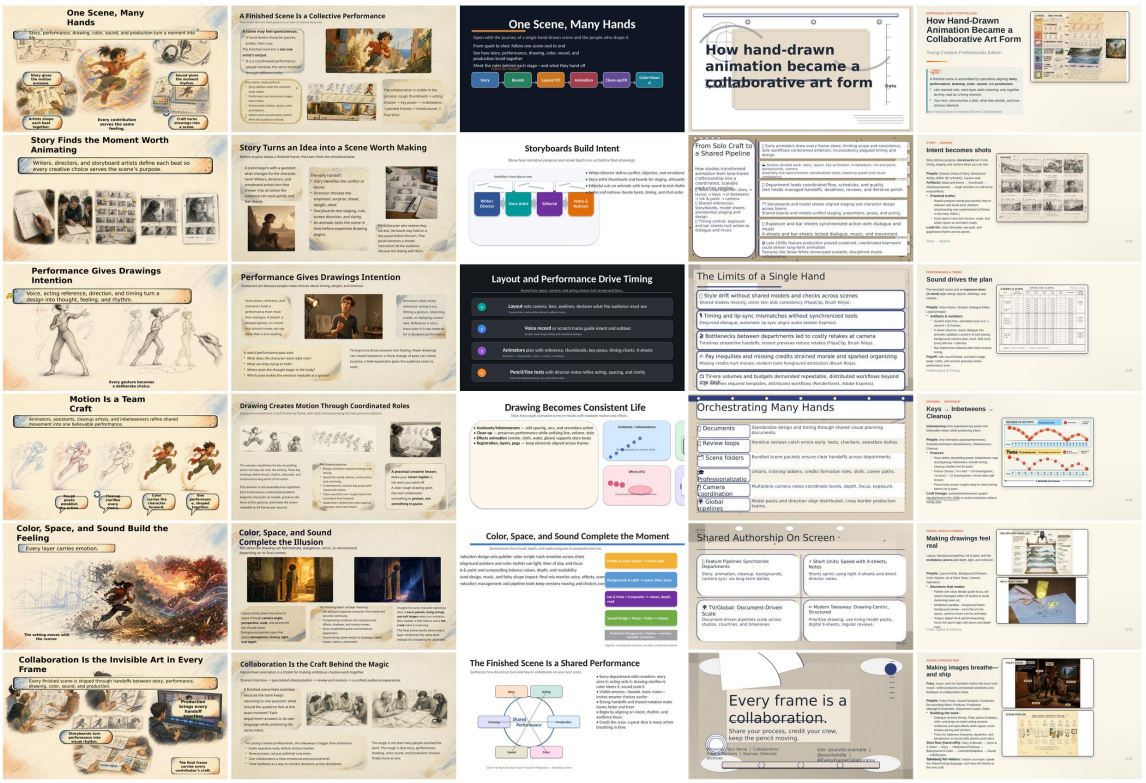


Figure 9: Fixed-combination comparison for "How hand-drawn animation became a collaborative art form." This six-slide narrative case uses GPT+GPT and requires a coherent sequence of creative stages.



Figure 10: Fixed-combination comparison for "How census data becomes local public funding." This goal-oriented case uses `Qwen+GPT` and combines a causal chain with an accessible public-information style.

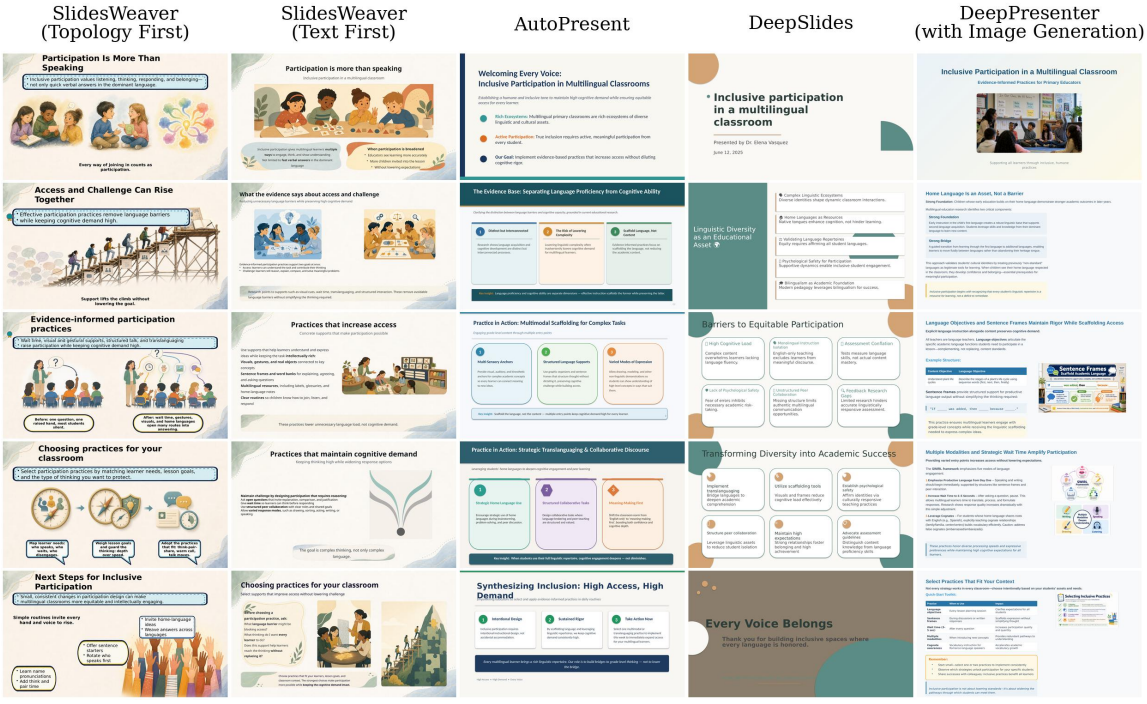


Figure 11: Fixed-combination comparison for "Inclusive participation in a multilingual classroom." This high-demand case uses `Qwen+GPT` and must coordinate access, participation, and cognitive challenge.

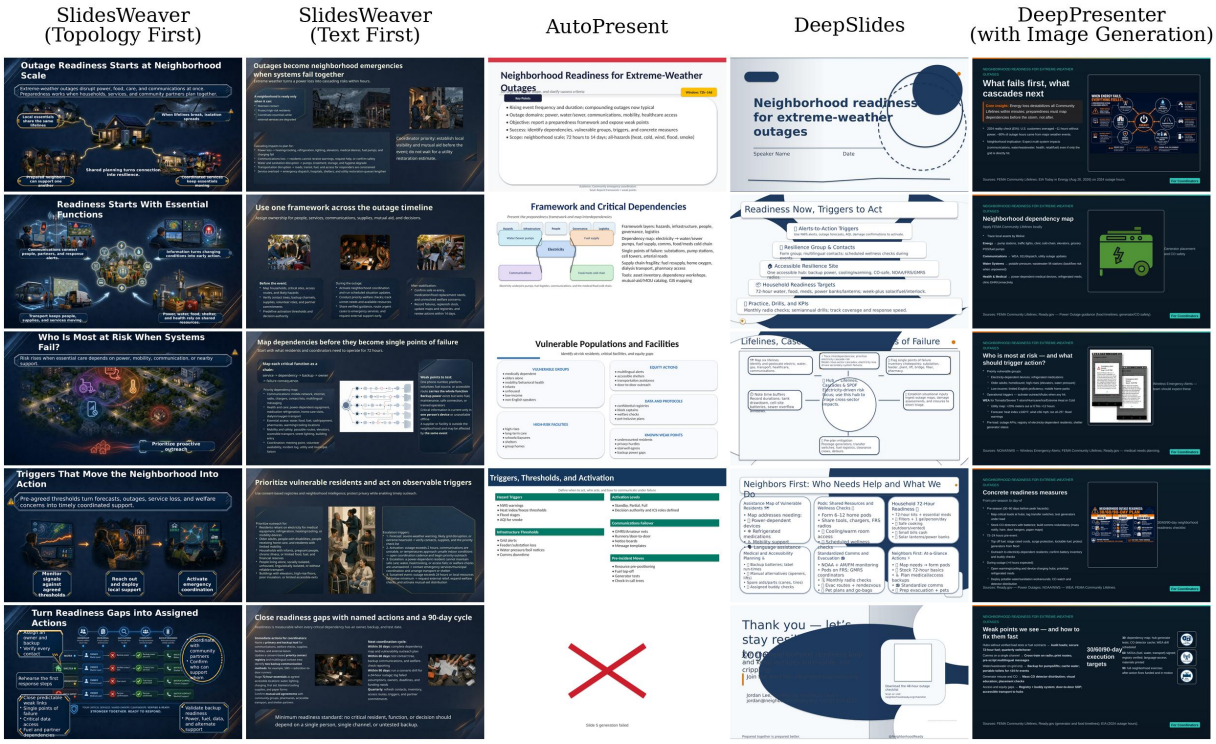


Figure 12: Fixed-combination comparison for "Neighborhood readiness for extreme-weather outages." This high-demand case uses GPT+GPT and requires dependencies, vulnerable groups, triggers, and concrete readiness actions to remain visible together.